

Comparative table showing how key **database, table, record, field, and HA/DR terms** differ across **SQL Server, MongoDB, PostgreSQL, and Oracle**.

Database Terminology Comparison

Concept / Function	SQL Server	MongoDB	PostgreSQL	Oracle
Database	Database	Database	Database	Database (or Schema within Database)
Table / Collection	Table	Collection	Table	Table
Row / Record	Row	Document	Row	Row
Column / Field	Column	Field	Column	Column
Primary Key	Primary Key	_id (unique per document)	Primary Key	Primary Key
Foreign Key	Foreign Key	Manual Reference (No enforced FK)	Foreign Key	Foreign Key
Schema (Logical Grouping)	Schema	N/A (Each DB = implicit schema)	Schema	Schema
Index	Index	Index	Index	Index
View	View	View (read-only aggregation pipeline view)	View	View
Stored Procedure	Stored Procedure	Function (written in JS or app side)	Stored Procedure / Function	Stored Procedure / Function
Trigger	Trigger	Change Stream / Trigger (via Realm or app layer)	Trigger	Trigger
Transaction	Transaction	Multi-document transaction (since 4.0)	Transaction	Transaction
Backup	Full / Differential / Log Backup	mongodump / mongorestore or Snapshots	pg_dump / Base backup	RMAN (Recovery Manager)
Restore	RESTORE DATABASE	mongorestore	pg_restore	RMAN RESTORE
User / Role Management	Login / User / Role	User / Role	Role / User	User / Role
Query Language	T-SQL	MQL (Mongo Query Language)	SQL (PostgreSQL dialect)	SQL (PL/SQL)
Execution Plan	Execution Plan	Explain Plan	EXPLAIN / ANALYZE	EXPLAIN PLAN
Data Type Example	INT, VARCHAR, DATETIME	int, string, date, array, object	INTEGER, TEXT, TIMESTAMP	NUMBER, VARCHAR2, DATE
Constraints	PK, FK, UNIQUE, CHECK	Schema validation (JSON schema)	PK, FK, UNIQUE, CHECK	PK, FK, UNIQUE, CHECK

 High Availability (HA) & Disaster Recovery (DR) Terminology

HA / DR Concept	SQL Server	MongoDB	PostgreSQL	Oracle
Replication Type	Transactional / Merge / Snapshot	Replica Set	Streaming Replication	Data Guard
Automatic Failover	Always On Availability Groups	Replica Set (Primary-Secondary)	Patroni / Repmgr / Native HA	Data Guard (Fast-Start Failover)
Synchronous Replication	Yes (Synchronous AG Mode)	Yes (Primary-Secondary Sync Option)	Yes (Synchronous Standby)	Yes (Synchronous Standby)
Asynchronous Replication	Yes	Yes	Yes	Yes
DR / Secondary Site	Log Shipping / Always On DR	Replica Set across DCs / Cluster	Physical / Logical Replication	Data Guard (Physical / Logical)
Cluster Management	Failover Cluster Instance (FCI) / WSFC	Sharded Cluster + Config Servers	Clustered via Patroni / Pgpool-II	Oracle RAC (Real Application Clusters)
Sharding / Partitioning	Table Partitioning	Sharded Cluster (key-based)	Table Partitioning / FDW / Sharding	Table Partitioning / RAC / Sharding
Load Balancing	Read-Only Routing (AG)	Read from Secondaries	Load Balancing via Pgpool / HAProxy	Services in RAC handle load
Monitoring Tools	SQL Server Management Studio (SSMS) / SQL Agent	MongoDB Atlas / Ops Manager	pgAdmin / pgbouncer / Prometheus	Enterprise Manager (OEM)
Backup & Restore HA	Always On + Backups	Replica Set + Snapshots	WAL Archiving + Replication	RMAN + Data Guard
Cluster Coordinator	Windows Failover Cluster	Config Servers + mongos	etcd / Consul (for HA frameworks)	CRS (Cluster Ready Services)

 Schema vs Schema-less Design

Aspect	SQL Server	MongoDB	PostgreSQL	Oracle
Schema Enforcement	Strict (Defined Columns)	Schema-less (Dynamic JSON)	Strict (but supports JSONB)	Strict
Flexibility	Moderate	Very High	Moderate to High (JSONB support)	Moderate
Best Use Case	OLTP, Analytics	Flexible / Unstructured / Scalable Data	OLTP / OLAP / Hybrid	Enterprise OLTP / Data Warehouse

 Performance, Scaling, and Query Optimization Terminology Comparison

Concept / Function	SQL Server	MongoDB	PostgreSQL	Oracle
Query Optimizer	Cost-Based Optimizer (CBO)	Query Planner / Optimizer	Cost-Based Optimizer (CBO)	Cost-Based Optimizer (CBO)
Execution Plan	SET SHOWPLAN / Actual Execution Plan	explain() / explain("executionStats")	EXPLAIN / EXPLAIN ANALYZE	EXPLAIN PLAN FOR
Index Types	B-tree, Clustered, Non-Clustered, Columnstore, Full-Text	B-tree, Compound, Multikey, Text, Geospatial	B-tree, Hash, GIN, GiST, BRIN	B-tree, Bitmap, Function-based, Clustered
Clustered Index	Supported (1 per table)	Not applicable	Supported (via physical order)	Supported (Index-organized table)
Non-Clustered Index	Yes	Yes	Yes	Yes
Covering Index	Yes (INCLUDE clause)	Yes (projection index)	Yes	Yes
Partitioning	Range, Hash, List, Composite	Sharding by key or hash	Range, List, Hash, Composite	Range, Hash, List, Interval, Composite
Parallel Query Execution	Yes (parallel scans, joins)	Limited (depends on sharding)	Yes (parallel query and workers)	Yes (parallel execution framework)
Materialized View	Indexed View	Materialized View	Materialized View	Materialized View
Caching	Buffer Pool, Plan Cache	WiredTiger Cache	Shared Buffers, Query Cache	Buffer Cache, Result Cache
In-memory Option	Memory-Optimized Tables	In-memory Storage Engine	In-memory Tables / UNLOGGED Tables	In-memory Option (Oracle In-Memory)
Compression	Row / Page Compression	WiredTiger Compression (Snappy, Zstd)	TOAST Compression	Advanced Compression Option
Statistics & Histograms	Auto Update Statistics	Query Planner Stats	ANALYZE Command	DBMS_STATS Package
Connection Pooling	Managed by SQL Client / Connection Pool	Managed by Driver / mongos	PgBouncer / Pgpool-II	Oracle Connection Manager / UCP
Concurrency Control	Lock-based + MVCC	MVCC (Optimistic)	MVCC	MVCC
Isolation Levels	READ COMMITTED, REPEATABLE READ, SERIALIZABLE, SNAPSHOT	Read Concern: local, majority, linearizable	READ COMMITTED, REPEATABLE READ, SERIALIZABLE	READ COMMITTED, SERIALIZABLE
Locking Mechanism	Page / Row / Table Locks	Document-level Lock	Row-level Lock	Row-level Lock
Query Hinting	Yes (OPTION (FORCESEEK), etc.)	Yes (\$hint)	Yes (SET enable_*)	Yes (/+ HINT */)

Concept / Function	SQL Server	MongoDB	PostgreSQL	Oracle
Monitoring / Profiling Tools	SSMS, Extended Events, DMV, PerfMon	Profiler, mongotop, mongostat, Atlas Metrics	pg_stat_activity, EXPLAIN, pgBadger	Enterprise Manager (OEM), AWR, ASH, ADDM
Load Testing Tools	SQLQueryStress, HammerDB	YCSB, JMeter	pgBench	Swingbench, HammerDB

Scalability & Deployment Patterns

Scaling Type	SQL Server	MongoDB	PostgreSQL	Oracle
Vertical Scaling	Add CPU / RAM	Add CPU / RAM	Add CPU / RAM	Add CPU / RAM
Horizontal Scaling	Limited (Federation, Elastic Query)	Sharding (Native)	Citus / FDW / Sharding	Sharding / RAC
Read Scaling	Read-only replicas (AG Secondaries)	Secondary Reads	Streaming Replicas	Active Data Guard Read-Only Standby
Write Scaling	Primary node	Primary node (single writer per shard)	Primary node	Primary or Multi-Master (RAC)
Multi-Region Deployment	Always On (Geo-Replica)	Global Clusters	Logical Replication / Cloud SQL Cross-Region	GoldenGate / Data Guard
Elastic Scaling (Cloud)	Azure SQL Elastic Pools	Atlas Auto-Scaling	Cloud SQL / Aurora PG	Autonomous DB Auto-Scaling
Cloud Native Service	Azure SQL Database	MongoDB Atlas	Cloud SQL / AWS Aurora / Azure PG	Oracle Autonomous Database

Summary Insights

Category	SQL Server	MongoDB	PostgreSQL	Oracle
Best For	Enterprise OLTP + BI + Data Warehousing	Flexible JSON-based, Scalable Apps	Open-source Enterprise / Hybrid	Mission-critical OLTP + Data Warehouse
Ease of Setup	Moderate	Easy	Moderate	Complex
License / Cost	Proprietary (per core)	Free (Community), Paid (Atlas)	Open Source	Proprietary (expensive)
Learning Curve	Moderate	Easy (JSON/NoSQL model)	Moderate	Steep
Cloud Friendliness	Azure SQL Native	Atlas Multi-cloud	Cloud-native integrations	Oracle Cloud (Autonomous)